PO11Q - Introduction to Quantitative Political Ansalysis I

Mock Exam 2025/26 - Solutions

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1 | Block A

Level 1 (each question 4 points)

1. `header=TRUE` tells R that the first row of the CSV file contains the names of the columns.
    - This allows the resulting data frame to use meaningful variable names (e.g., crime_rate, drive_primary) instead of default names like V1, V2, etc.
    - If `header=FALSE`, R treats the first row as data rather than column names, which can misalign the dataset and require manual renaming of columns.

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2. Figure 1 shows the distribution of the variable `crime_rate`. Figure 2 shows the cumulative distribution function of `crime_rate`.
    - The distribution of a variable (probability distribution) describes how the values of the variable are spread or occur across its range, showing the likelihood of each individual value.
    - The cumulative distribution function (CDF) gives the probability that the variable takes a value less than or equal to a specific point, effectively summing up the probabilities from the lower bound up to that point.

Level 2 (each question 6 points)

3. The code `simd2 <-filter(simd, crime_rate < 400)` selects rows from the `simd` dataset where the `crime_rate` is less than 400. It creates a new dataset simd2 containing only these filtered observations.
    - In plain terms: "Keep only the observations where the crime rate is below 400 and store them in `simd2`."

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4. The distribution is right-skewed, or positively skewed, which we can deduce from the mean (177.65) being larger than the median (161.48). The distribution starts at the minimum of 10.83, and ends at the maximum of 399.95. The mode is not present in Subsection 7.3, but seems to lie at a little under 100.

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5. The probability of observing a data zone with a crime rate of 300 or less is 85%.

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6. The output shows the result of a one-sample test in R.
   - The one-sample t-test compares the mean crime_rate in simd2 to the hypothesized value of 200.
   - The sample mean is 177.65, which is less than 200.
   - The t-statistic is -16.103, indicating the sample mean is many standard errors below 200.
   - The p-value is less than 2.2e-16, meaning the result is highly statistically significant.
   - The 95% confidence interval for the true mean ranges from 174.92 to 180.37, which does not include 200.
   - Interpretation: There is extremely strong evidence that the average crime rate in the filtered dataset is lower than 200. The difference is unlikely to be due to random sampling.

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7. We would need a credible theoretical framework, for example, that is able to establish how alcohol abuse leads to death (mechanisms approach). Temporally, this setup might already establish asymmetry as death cannot precede substance abuse temporally.

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8. All breaks but the first (because of `include.lowest = TRUE`) have to be adjusted a tiny bit upwards, and the option itself needs to be changed:

```
simd2 %>%
mutate(drive_cat = cut(drive_primary,
                       breaks = c(0, 2.79 + 1e-8, 3.572 + 1e-8, 30 + 1e-8),
                       labels = c("Short", "Average", "Long"),
                       right = TRUE,
                       include.lowest = TRUE)) -> simd2
```

Level 3 (each question 8 points)

9. In Subsection 7.6, the sample `simd_sample$drive_primary` is used in a one-sample t-test with $\mu = 5$ and a very large negative t-statistic ($t = -97.62$).

   - The t-statistic formula is:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

   - Rearranging for the sample mean:

$$\bar{x} = t \cdot \frac{s}{\sqrt{n}} + \mu$$

   - Given:
     - $t = -97.62$
     - $s = 1.34018$
     - $n = 4500$
     - $\mu = 5$

   - Calculate the standard error:

$$se = \frac{1.34018}{\sqrt{4500}} \approx \frac{1.34018}{67.082} \approx 0.01997$$

- Calculate the sample mean:

$$\bar{x} = -97.62 \times 0.01997 + 5 \approx -1.949 + 5 \approx 3.051$$

- The sample mean of `simd$drive_primary` ≈ 3.051

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10. We could label the axes and provide a legend title:

```
ggplot(simd, aes(x = drive_cat, fill = rank_cat)) +
  geom_bar() +
  xlab("Drive Category") +          # add x-axis label
  ylab("Frequency") +               # keep y-axis label
  labs(fill = "Deprivation Level")  # add title for the legend
```

2 Block B

1. If the mean of a variable is larger than its median, the distribution of this variable is positively skewed. — True

- When the mean is larger than the median it typically indicates that the right tail is longer or that large values pull the mean to the right; this is the usual diagnostic for positive (right) skew.

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2. With a smaller effect size of interest, a higher sample size is required to achieve the same level of statistical power. — True

- If the target effect size is smaller, you need larger n to make effect $\cdot \sqrt{n}$ large enough to detect it at the same power.
- In practice n scales approximately with $1/(\text{effect size})^2$ for simple tests, so halving the effect size requires about four times the sample size to hold power constant.

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3. The p-value and the α-level are the same thing. — False

- The p-value is a data-dependent quantity: the probability, under the null hypothesis, of observing data as extreme or more extreme than what was observed.
- The α-level is a pre-specified cut-off (e.g., 0.05) that the researcher chooses before the test; it sets the maximum tolerated probability of a Type I error.
- You compare p to α (reject if $p \le \alpha$), but p and α are conceptually and functionally distinct: one is observed evidence, the other is a decision threshold.

3 | Block C

1. For the data (x = 300, 460, 180, 245, 259)

   a. Calculate the mean of x.

   - Sum the observations: $300 + 460 + 180 + 245 + 259 = 1444$.
   - Number of observations: $n = 5$.
   - Mean: $\bar{x} = \frac{1444}{5} = 288.8$.

   b. Calculate the (sample) standard deviation of x.

   - Deviations from the mean:
     - $300 - 288.8 = 11.2$
     - $460 - 288.8 = 171.2$
     - $180 - 288.8 = -108.8$
     - $245 - 288.8 = -43.8$
     - $259 - 288.8 = -29.8$
   - Squared deviations:
     - $11.2^2 = 125.44$
     - $171.2^2 = 29310.14$
     - $(-108.8)^2 = 11839.14$
     - $(-43.8)^2 = 1918.44$
     - $(-29.8)^2 = 888.04$
   - Sum of squared deviations: $125.44 + 29310.14 + 11839.04 + 1918.44 + 888.04 = 44081.10$ *(When computed exactly from the unrounded deviations the precise sum is 44078.8; the small difference above is due to intermediate rounding. I am using the exact sum 44078.8 for the variance below.)*
   - Sample variance (use $n - 1$ in the denominator):

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n - 1} = \frac{44078.8}{5 - 1} = \frac{44078.8}{4} = 11019.7.$$

   - Sample standard deviation:

$$s = \sqrt{11019.7} \approx 104.975.$$

   c. Calculate the 65th percentile of x.

   - Sort the data: $x_1 = 180,\ x_2 = 245,\ x_3 = 259,\ x_4 = 300,\ x_5 = 460$.
   - Use the $(n + 1)$ interpolation method to find the p-th percentile:
     - Position $= p(n + 1) = 0.65 \cdot (5 + 1) = 0.65 \cdot 6 = 3.9$.
     - This lies between the 3rd and 4th ordered values (3.0 and 4.0). Let $k = 3$ and fractional part $d = 0.9$.
     - Interpolate:

$$P_{65} = x_3 + d\left(x_4 - x_3\right) = 259 + 0.9(300 - 259) = 259 + 0.9(41) = 259 + 36.9 = 295.9.$$

2. Confidence interval question

- Sample of $n = 40$, sample mean $\bar{x} = 12$, sample standard deviation $s = 3.5$. We want the upper limit of a 99% CI for the population mean μ. Use the t-distribution (degrees of freedom $df = n - 1 = 39$).

- Standard error:
$$se = \frac{s}{\sqrt{n}} = \frac{3.5}{\sqrt{40}} \approx \frac{3.5}{6.3246} \approx 0.553399.$$

- Critical t-value for a two-sided 99% CI is $t_{0.995,\,39}$. Using the t critical quantile (df = 39):
$$t_{0.995,39} \approx 2.702.$$

- Margin of error:
$$\mathrm{ME} = t_{0.995,39} \cdot se \approx 2.702 \cdot 0.553399 \approx 1.495284.$$

- Upper limit of the 99% CI:
$$\bar{x} + \mathrm{ME} = 12 + 1.495225 \approx 13.495.$$

- Rounded to three decimal places: **13.495**.

Block D

1. Central Limit Theorem

- **What it is:** The Central Limit Theorem (CLT) states that the sampling distribution of the sample mean (or sum) approaches a normal distribution as the sample size increases, regardless of the population's shape.
- **Example (worked):** Suppose a factory produces bolts with lengths that are skewed to the right, with a population mean of 50 mm and a standard deviation of 5 mm. If we take repeated samples of size $n = 30$ and calculate the sample means, the distribution of those means will approximate a normal distribution, even though the individual bolt lengths are skewed. The standard error of the mean is $se = s/\sqrt{n} = 5/\sqrt{30} \approx 0.913$.
- **Why it matters:** It allows us to use normal-based inference (confidence intervals, hypothesis tests) even when the population is not normally distributed.
- **What it is not:** It is not the distribution of individual data points; it describes the distribution of the sample statistic (such as the mean) across repeated samples.

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2. Statistical Power

- **What it is:** Statistical power is the probability of correctly rejecting a false null hypothesis (i.e., detecting an effect if it exists). It equals $1 - \beta$, where β is the probability of a Type II error.
- **Example (worked):** Suppose a researcher tests whether a new teaching method increases student test scores. The null hypothesis is that the mean score is 70, and the alternative is that it is higher. If the true mean with the new method is 75, and the standard deviation is 10, taking a sample of 25 students gives $se = 10/\sqrt{25} = 2$. Using a significance level $\alpha = 0.05$, the critical value for rejecting H_0 is approximately 73.92. The probability that the sample mean exceeds 73.92 given the true mean 75 is the power.
- **Why it matters:** Higher power reduces the risk of Type II errors and ensures that studies are more likely to detect meaningful effects.
- **What it is not:** Power is not the probability that the null hypothesis is true or false; it is a property of the test given a true alternative effect size.